Exponential graphs

1

a Yes **b** Yes **c** No **d** No

e No **f** No

2 Graph B

3

a Graph R

b For $x < 0$, the graph of $y = (5)^x$ is below the graph of $y = (3)^x$. This is because negative values of x result in fractional function values, and for any negative value of x, $(5)^x$ will result in a smaller value than $(3)^x$.

4

a Increasing **b** Decreasing

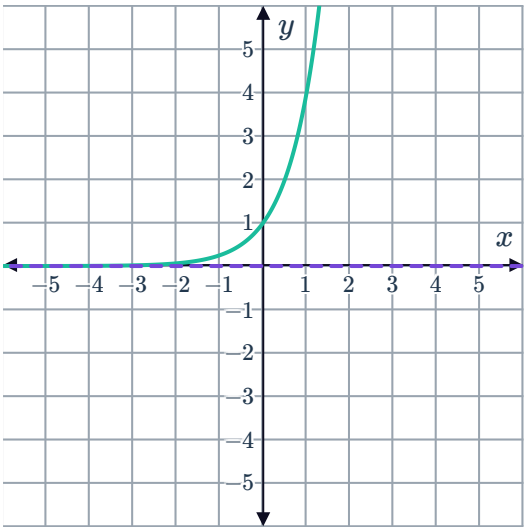
Increasing exponential graphs

5

a

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{64}$	$\frac{1}{16}$	$\frac{1}{4}$	1	4	16	64

b



c As x increases, the y-values also increase.

d As x decreases, the y-values also decrease.

e No

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f $y = 1$

g $x \approx 0.8$

6

a

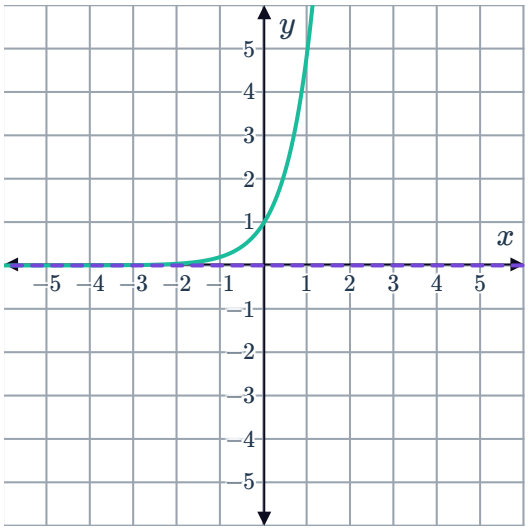
i 1

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{64}$	$\frac{1}{16}$	$\frac{1}{4}$	1	4	16	64

iii $y = 0$

iv



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b

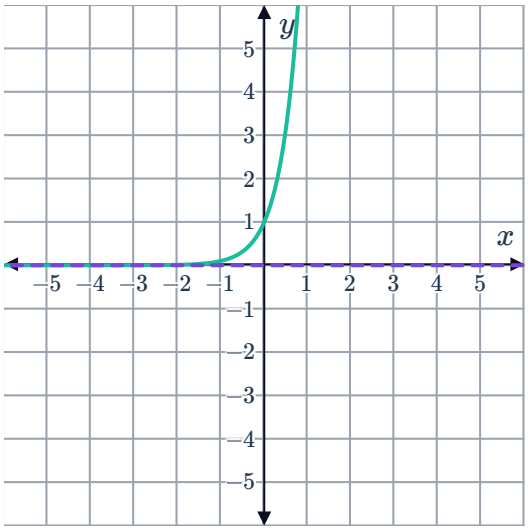
i $y = 1$

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{1\,000}$	$\frac{1}{100}$	$\frac{1}{10}$	1	10	100	$1\,000$

iii $y = 0$

iv



c

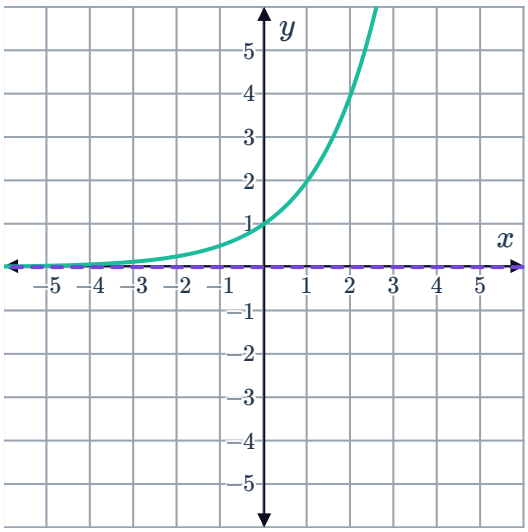
i $y = 1$

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

iii $y = 0$

iv



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7

a No

b No

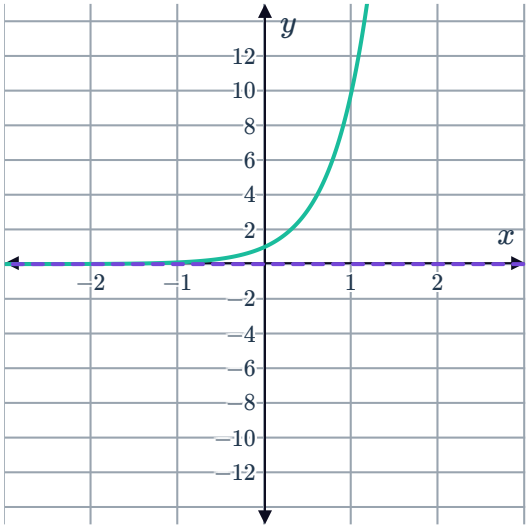
c ∞

d 0

e $y = 1$

f 0

g



h $x \approx 0.4$

8

a

i $y = 0$

ii $y = 0$

b

i

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27

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g $x \approx -2.2$

11

a

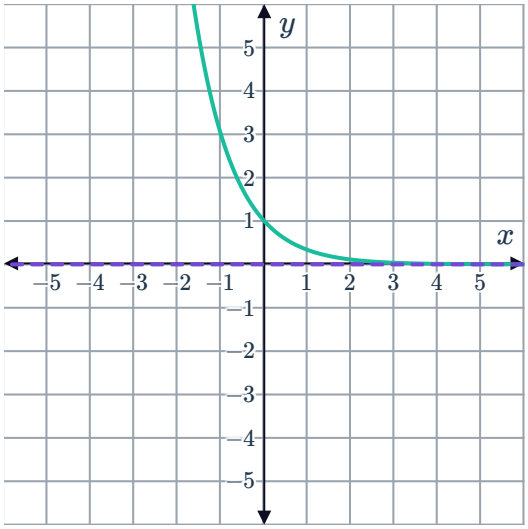
i $y = 1$

ii

x	-3	-2	-1	0	1	2	3
y	27	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{27}$

iii $y = 0$

iv



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b

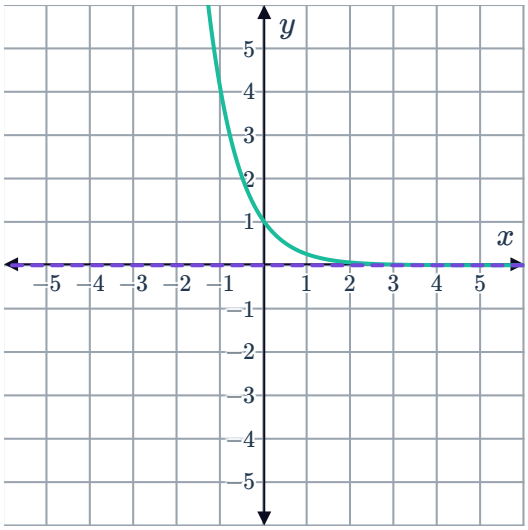
i $y = 1$

ii

x	-3	-2	-1	0	1	2	3
y	64	16	4	1	$\frac{1}{4}$	$\frac{1}{16}$	$\frac{1}{64}$

iii $y = 0$

iv



c

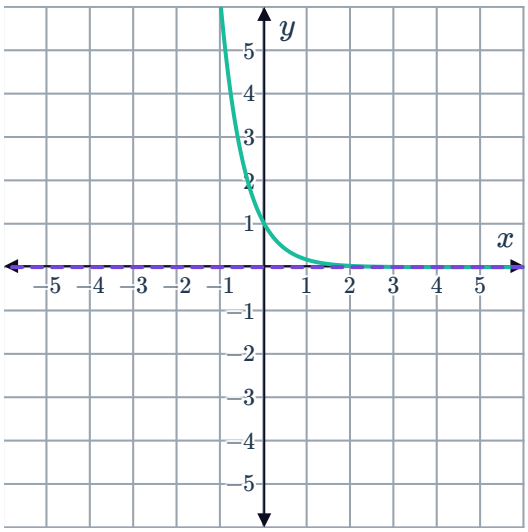
i $y = 1$

ii

x	-3	-2	-1	0	1	2	3
y	216	36	6	1	$\frac{1}{6}$	$\frac{1}{36}$	$\frac{1}{216}$

iii $y = 0$

iv



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12

a No

b No

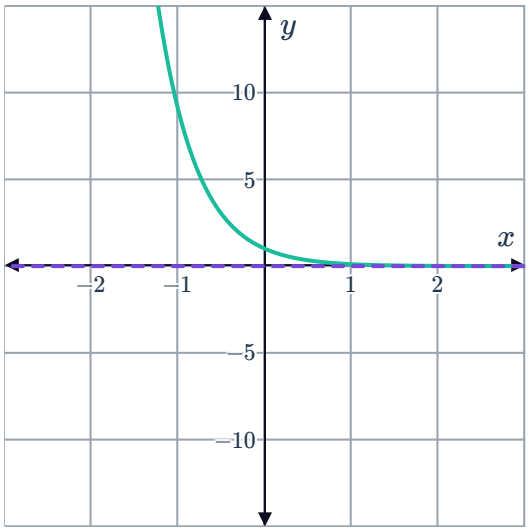
c 0

d ∞

e $y = 1$

f 0

g



h $x \approx -0.7$

13

a

i $y = 0$

ii $y = 0$

b

i

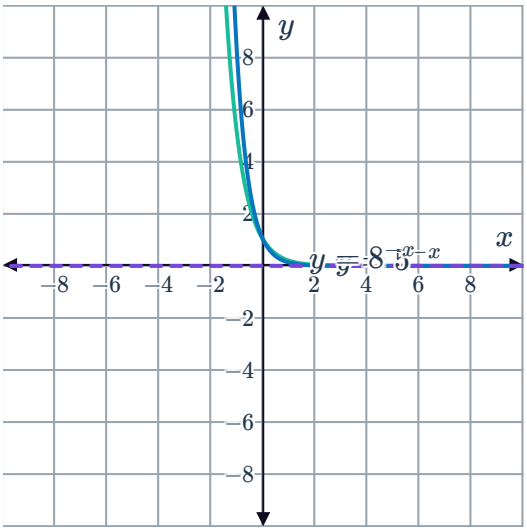
x	-3	-2	-1	0	1	2	3
y	125	25	5	1	$\frac{1}{5}$	$\frac{1}{25}$	$\frac{1}{125}$

ii

x	-3	-2	-1	0	1	2	3
y				1	$\frac{1}{8}$	$\frac{1}{64}$	$\frac{1}{512}$

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c



d $(0, 1)$

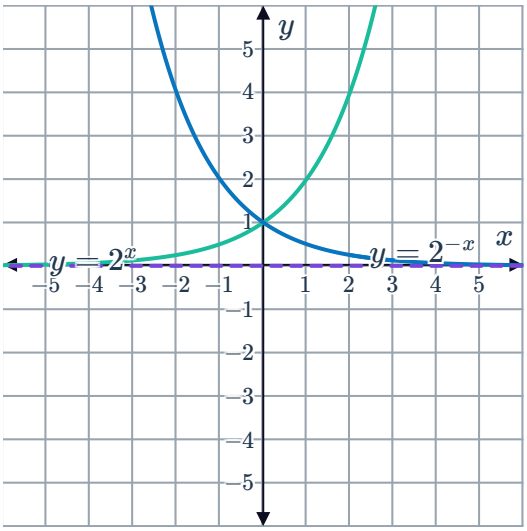
Dilations and reflections

14 By reflecting the graph about the y-axis.

15

a A reflection about the y-axis.

b

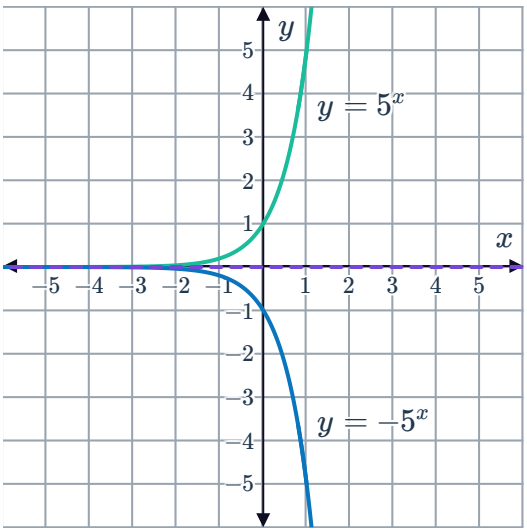


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16

a A reflection about the x-axis.

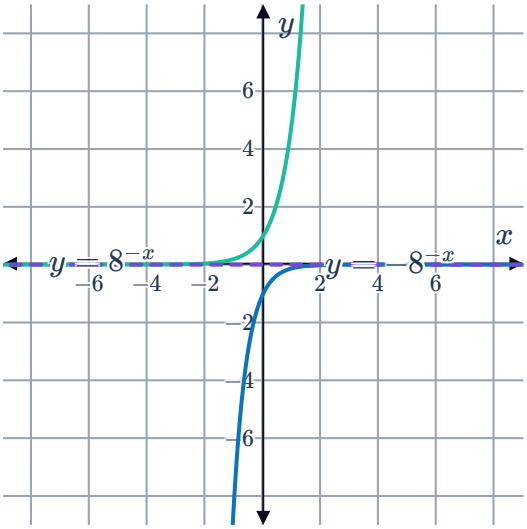
b



17

a A reflection about both the x-axis and y-axis.

b

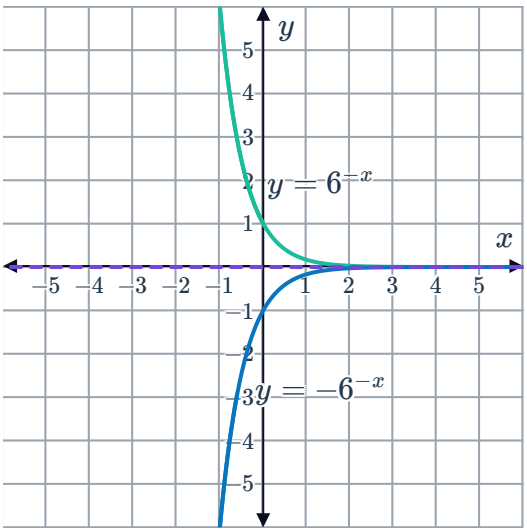


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18

a A reflection about the x-axis.

b



c As x increases, the graph of $y = -6^{-x}$ is increasing at a decreasing rate.

19 $y = 4^x$

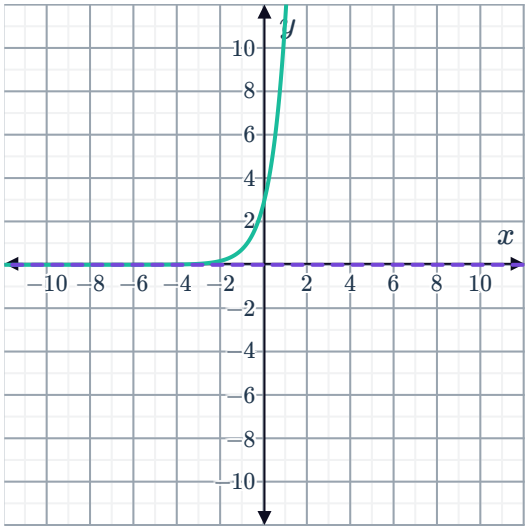
20

a $y = 3$

b No

c ∞

d



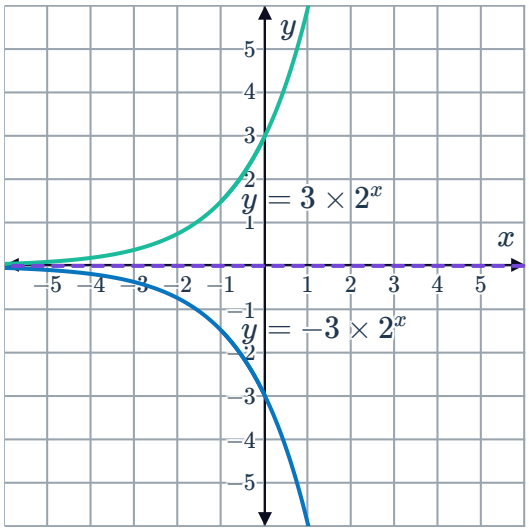
21

a $y = 3$

b A reflection about x-axis.

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c



d $y = 3 \times 2^x$

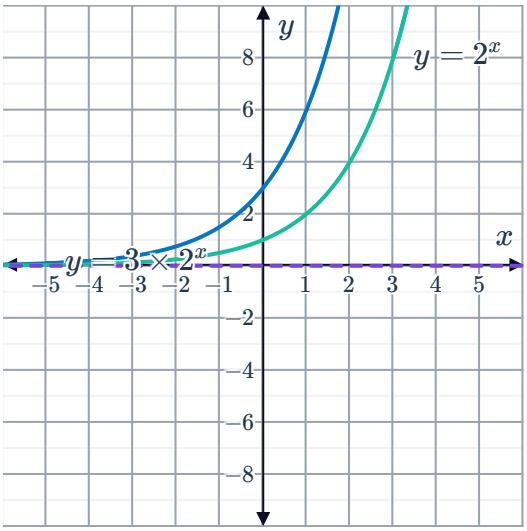
22

a

Point on original graph	$\left(-1, \frac{1}{2}\right)$	$(0, 1)$	$(1, 2)$	$(2, 4)$
Point on new graph	$\left(-1, \frac{3}{2}\right)$	$(0, 3)$	$(1, 6)$	$(2, 12)$

b $y = 3 \times 2^x$

c



d Above

e Above

Translations

23

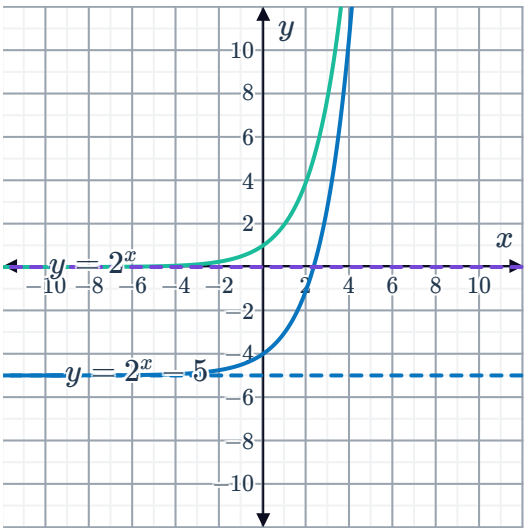
a $y = 1$ **b** $y = 4$ **c** $y = 0$ **d** $y = 3$

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24

a Move the graph downwards by 5 units.

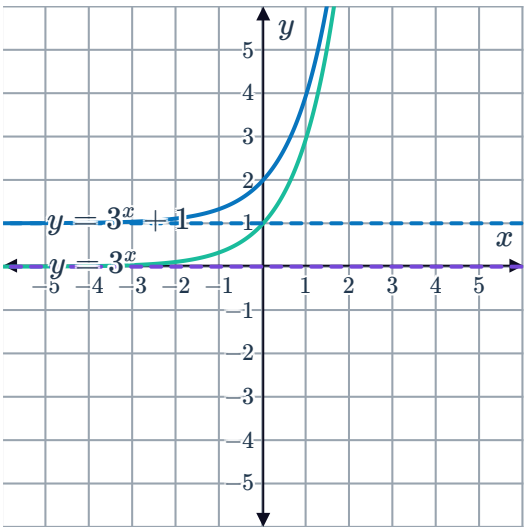
b



25

a Move the graph upwards by 1 unit.

b

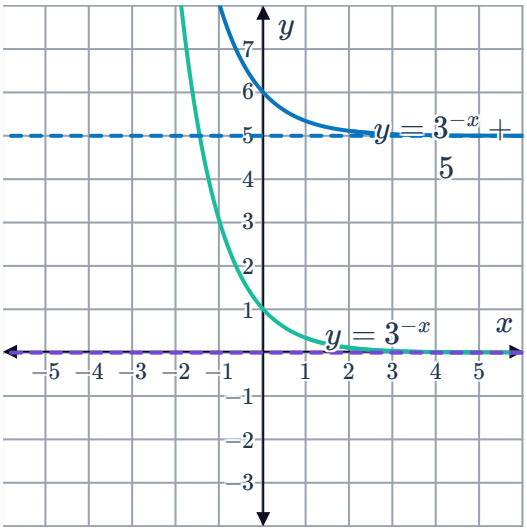


26

a Move the graph upwards by 5 units.

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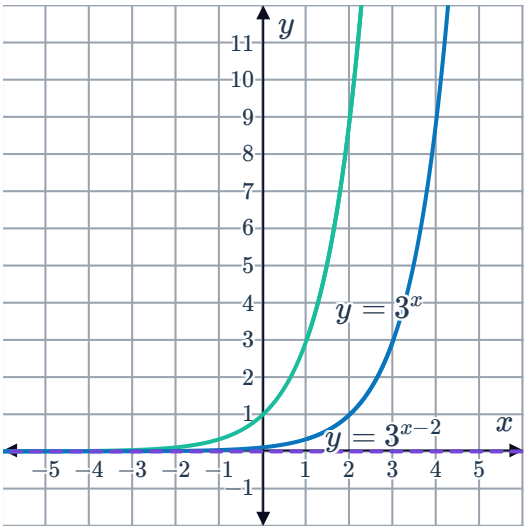
b



27

a Move the graph to the right by 2 units.

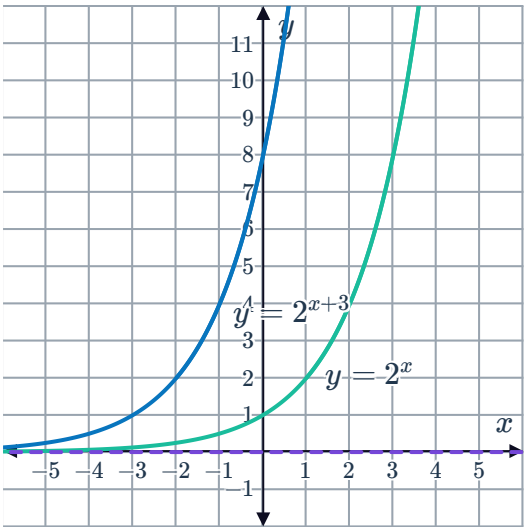
b



28

a Move the graph to the left by 3 units.

b

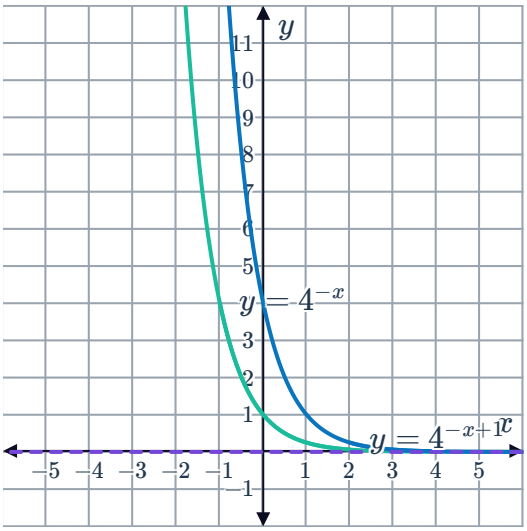


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29

a Move the graph to the left by 1 units.

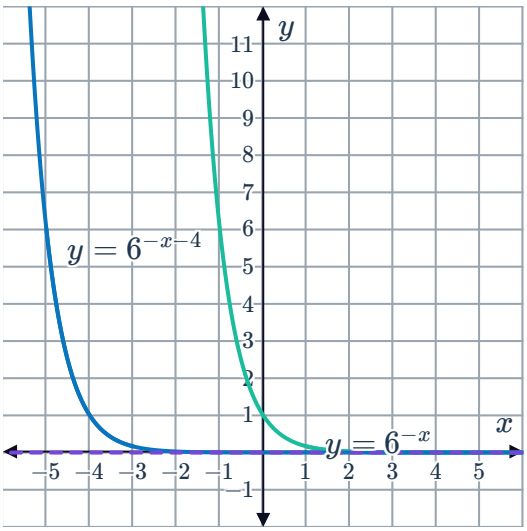
b



30

a Move the graph to the right by 4 units.

b



31 $y = 2^x - 9$

32 $y = 12^{x+6}$

33 $y = 3^{-x-2} - 7$

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